Multiphysic Simulation Approach for 5G Filter Design

Domenico Loricchio
Ansys Inc.
Milan, Italy
domenico.loricchio@ansys.com

Alberto Di Maria
Ansys Inc.
Milan, Italy
alberto.dimaria@ansys.com

Samuel Lopez-Ruiz
Ansys, Inc.
Madrid, Spain
samuel.lopez@ansys.com

Massimo Capodiferro
Ansys Inc.
Milan, Italy
massimo.capodiferro@ansys.com

***Abstract*— Modern electronic devices are subjected to increasing power levels, higher frequencies, and sometime to a miniaturization process. All these aspects can lead to thermal and mechanical stress compromising safety, functionality, and reliability of our devices. The implementation of new technologies makes the use of predictive modeling techniques even more essential in designing electronic products. The aim of this paper is to describe and provide an automated multiphysic approach dedicated to 5G filter design. This simulation workflow of thermo-electro-mechanical stress effects within filters allows engineers to prevent failures during the early stages of the design process and to reduce the time-to-market. All the steps of the finite element analysis and the automated process have been performed with Ansys tools and PyAEDT library.**

Keywords—5G, filter, simulation, automation, reliability

I. Introduction

Driven by consumer demand for devices that extend the connected lifestyle, provide enhanced safety, and reduce environmental impact, electronic market is surging. Engineers are developing new technologies to satisfy these customer needs and to make the "dream" come true. Before every innovation sees the light of day, it is needed a rigorous analysis of each physical aspect that can modify the correct functioning of a device.

The authors would like to show all the steps used in this automated multiphysic simulation workflow for 5G filters design. Section II has been dedicated to geometry preparation of the mechanical CAD. This step is essential for the entire FEA (Finite Element Analysis) analysis described in the next sections and it will be highlighted how it is easy to automate this process reducing time dedicated to geometry pre-processing. Section III and IV describe electro-thermal and structural analysis, respectively. In these sections, authors present some new features available to make easier data exchange among several Ansys tools. Finally, in section V authors share some detail on automation of the entire flow and a comparison between simulation results achieved at the end of the multiphysic analysis and some measurement.

II. Geometry Preparation

The first step in the filter multiphysic analysis is to import the filter model. The geometry preparation phase performs a series of steps aiming to guarantee that the model is suitable for electromagnetic, thermal, and structural simulations. In addition, as the CAD model is usually shared in step file format, some checks and healing must be performed during the import. Finally, some optimizations can be done in order to speed up the computational time of the simulation.

This phase is done in Ansys SpaceClaim due to its possibility to be entirely scripted in Python and its native compatibility with all other Ansys tools. In the following there is a detail of the main steps performed.

A. Import geometry, check for errors and healing

First, the step file is imported. Upon import the faces can be automatically connected and solids closed. Right after the import the geometry is checked for errors and an automatic healing is applied. Although the automatic heling cannot automatically fix all possible errors, it is powerful enough to fix all the issues generated by the import operation (i.e. translation between CAD formats).

B. Organize bodies and components to be compatible with HFSS

Ansys HFSS modeler is somehow limited respect to a full-fledged mechanical model editor, so it does not support linked bodies and nested components. The bodies and components need to be reorganized and unlinked to be correctly imported in Ansys HFSS.

C. Solve interferences and check for isolated objects

While EM simulators are generally tolerant respect to interferences between bodies, the same it is not true for a structural simulator. As the same space cannot be physically occupied by two objects, the mechanical simulation would fail. The interferences are automatically solved based on specific precedence rules set by the user (e.g., volume or material). Similarly, an object floating in the vacuum can be normal for an

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EM simulation but must be detected before launching a mechanical simulation.

D. *Internal volume and internal objects detection*

To correctly evaluate and apply the Heat Transfer Coefficients (HTC), the internal and external surfaces (and consequently the internal objects) must be detected. In the geometry preparation phase, a sophisticated algorithm is used to extract the internal volume automatically, without user intervention. Usual algorithms require the user to select (seed) a surface, to give a hint of which part should be considered internal and which one external. The developed algorithm does the same without user intervention.

E. *Chassis and cover detection*

The chassis and the cover of the model are also automatically detected. This information is used to apply mechanical constraints to the significant bodies.

F. *Screw simplifications*

To speed up the simulation time, it is also possible to perform some simplifications. The most common is the "screw simplification" where all screws details, like head shape, threads, and tapering are removed.

G. *Rounding simplification*

Another simplification that is possible to perform is the rounding removal. Rounded edges are usually a mechanical detail that is irrelevant for the EM simulation, yet a lot of mesh elements are required to correctly capture the curvature of the rounding.

III. Electro-Thermal Analysis

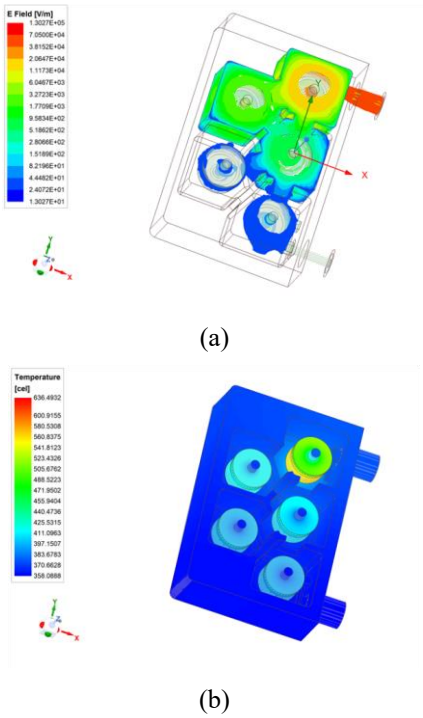
Electro-thermal analysis was conducted in AEDT (Ansys Electronics Desktop) using Ansys HFSS for the electromagnetic simulation and Ansys Icepak for the thermal one (Fig. 1).

These two tools have the possibility to exchange information within AEDT in an easy way. Solving electromagnetic analysis, it is possible to import power distribution coming from HFSS as EM losses in Icepak and when the thermal simulation is performed this information is translated in temperature distribution.

A. *Electromagnetic Simulation*

Ansys HFSS is the best-in-class 3D high frequency electromagnetic simulation software for designing and simulating high-frequency electronic products such as antennas, antenna arrays, RF or microwave components, high-speed interconnects, filters, connectors, IC packages and printed circuit boards.

In earlier releases, this software introduced new meshing technologies that are very useful for microwave filter and broadband antennas. The automatic adaptive meshing procedure at a single frequency is the cornerstone technology that enables Ansys HFSS to calculate the electric fields accurately and efficiently. In most cases, the use of the single solution frequency is sufficient to generate accurate results. For broadband, multi-band, multi-resonant devices it is useful to perform adaptive meshing at multiple frequencies. The multi-frequency adaptive meshing process is similar to the single one; the difference is that the single mesh describing the geometry of



(a)

(b)

Fig. 1 - Electro-thermal analysis performed in AEDT (a) E-field obtained with Ansys HFSS. (b) Temperature distribution obtained with Ansys Icepak.

the model is solved at multiple frequencies in parallel. For each frequency, the solutions are analyzed and Ansys HFSS determines automatically where to refine the mesh using information from the multiple frequencies to automatically reduce errors to the solution. The mesh is then refined locally, both in size and order of basis function, in the locations where the errors were relatively large. This refinement yields to a mesh that is more accurate and reliable at the frequencies used in the adaptive process. In this multiphysic flow it is possible to use both meshing techniques. Both approaches are able to provide good results for 5G cavity filter.

Another important aspect of the electromagnetic simulation is the type of ports chosen to excite filters. This decision could be critical in particular if we have to consider the possibility to run another electromagnetic simulation at the end of the multiphysic flow with a deformed mesh coming from structural analysis. Paying attention to port settings it is possible to excite cavity filters with wave ports and for other kind of structure as ceramic filter installed on carrier board or planar filter it is possible to also use circuit port. This allows the user to run electromagnetic simulation without problem also with deformed mesh coming from structural analysis. Most of the time, engineers do not have a filter CAD ready to use in a multiphysic analysis. For this reason, filter synthesis plays a key role in filter design process. In Ansys portfolio there is a solution that can help engineer during this design stage: Ansys Nuhertz Filter Solution. This tool can synthesize planar and digital filters and export a CAD in HFSS to run a FEM analysis. For synthesis of other kind of filters there is also the possibility to use a partner software called SynMatrix. This tool is a design platform for RF engineers and technicians that can help accelerate the microwave component design and manufacturing process. The typical RF/Microwave design process is very time-consuming

and expensive. This tool can design 3D filters as cavity, coax, dielectric resonators and optimize them using artificial intelligence. Also, in this case it is possible to simulate synthesized structures using Ansys HFSS to get accuracy and reliability of a FEM solver.

At the end of the electromagnetic simulation, it is possible to import the EM losses from HFSS to Icepak. We could compare EM loss mapping between these two tools through the volumetric loss density expression:

$$\rho = \frac{1}{2} Re\left(E \cdot \tilde{J} + j\omega B \cdot \tilde{H}\right) = \frac{1}{2} Re\left(E \cdot \tilde{J} - \nabla \times E \cdot \tilde{H}\right) \quad (1)$$

Where:

- E is the electric field
- $\tilde{J}$ is the conjugate of the volumetric current density
- B is the magnetic flux density
- $\tilde{H}$ is the magnetic field

Integrating (1) over the volume we obtain the losses in all the volumes that are part of the filter structure:

$$\rho_V = \int_V \rho \, dV \quad (2)$$

It is possible to evaluate this integral in HFSS.

B. *Thermal Simulation*

The EM losses coming from the electromagnetic simulation are then mapped in Ansys Icepak. It supports both surface and volumetric mapping. This thermal solver is now completely integrated in the same simulation environment, AEDT. It makes very easy the data exchange between these two tools. Thanks to this interaction it is possible to consider both 1- or 2-way coupling during the electro-thermal simulation. With 2-way coupling it is possible to evaluate the impact of temperature distribution on results in HFSS after Icepak simulation is completed. Both 1- and 2-way coupling need an accurate description of material properties that must be temperature dependent. For this reason, materials used in 3D Designs may have a thermal modifier assigned to change specific material properties with temperature. Thermal modifiers are multiplicative in nature and assigning them introduces spatial dependence. From theory [1] we get that temperature dependence of material resistivity is given by:

$$\sigma(T) = \sigma_0[1 + \alpha(T - T_{REF})] \quad (3)$$

Where:

- $\sigma(T)$ is resistivity of materials temperature dependent,
- T_{REF} is reference temperature, typically 22°C,
- σ_0 is resistivity at reference temperature,
- α is temperature coefficient of resistivity.

In Ansys Icepak it is possible to assign a thermal modifier for bulk conductivity of metals and considering that bulk conductivity is the reciprocal of electrical resistivity, it is easy to demonstrate that thermal modifier that must be considered for bulk conductivity of a metals in Ansys Icepak is equal to the reciprocal of (3).

Icepak is used to evaluate an accurate heat transfer coefficient (HTC) for the filter structures. It uses a Computational Fluid Dynamics (CFD) based thermal/flow analysis to account for all the modes of heat transfer and once computed, the HTC value is used downstream in the thermal mechanical calculations. A laminar flow regime has been considered in this workflow.

IV. Structural Analysis

Ansys Workbench [2] provides the ability to link electromagnetic to thermal and structural analysis as shown in Fig. 2.

Electromagnetic analysis (HFSS component in Fig. 2) is imported and linked to the Thermal Structural component, this link will load the power losses to perform the Structural Thermal simulation. The Structural Thermal setup needs an accurate boundary definition: convection and radiation. The convection values for each object of the device will be extracted from the previous step (Thermal Analysis) through HTC. Another approach can be taken, the temperature map from the Electro-Thermal analysis can be imported as an external data and linked directly to Static Structural Analysis. Both approaches give similar outcomes.

The goal of the wrokflow is to evaluate the deformations, due the thermal load acting on the device. This approach can also include filters with integrated PCB (i.e. Ceramic Filters), thanks to the Trace Mapping Technology [3]. It utilizes straightforward and efficient discretization methods, in combination with an innovative approach to material mapping, to represent the complex geometric nature of the ECAD components.

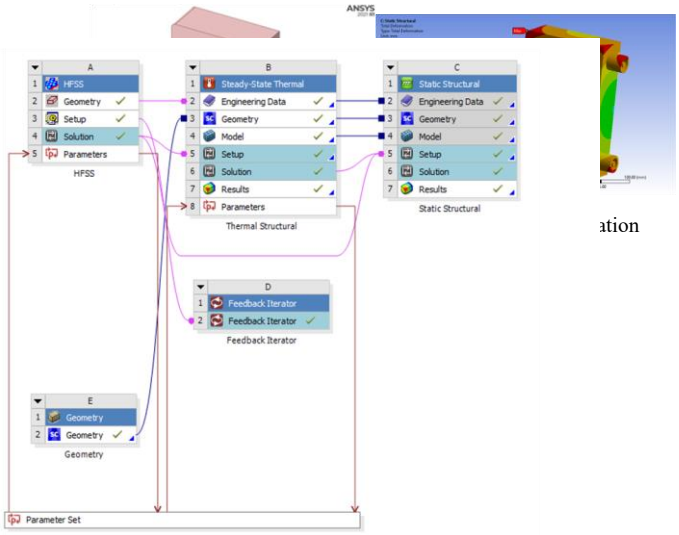


Fig. 2 - Ansys Workbench Analysis Schematic with Thermal Structural calculation

Materials properties are critical for an accurate simulation, using the right materials reduces the risks and cost of the designs. Properties such as thermal conductivity or thermal

expansion must be accurate to get the precision needed for a multiphysic simulation.

We used Ansys Granta Selector [4] which provides an extensive material selection database for electromagnetic, thermal, and structural simulation trusted by engineers and material experts.

Finally, to achieve very accurate results, the flow described in Fig. 2 must be 'Two-way', it means that the temperature and structural results must go back to the Electromagnetic simulation and this flow must be repeated iteratively until the temperature and deformation converge. This is automatically performed thanks to Ansys Workbench platform and the Feedback Iterator component (Fig. 2).

V. Automated Process & Results

Described simulation workflow can be fully automated using Python through the interconnectivity of the different Ansys tools and the new open-source package pyAEDT [5], which is a Python library that interacts directly with the AEDT API to make scripting simpler for the end user.

A. Automated process

The automated multiphysic workflow performed by the authors consists in 7 main steps:

1. Read CAD Model.
2. Assign Material Properties to each object.
3. Perform geometry preparation operation (Described in II).
4. Setup and Run HFSS (Described in III.A).
5. Setup and Run Icepak, exporting the HTC table (Described in III.B).
6. Setup and Run Workbench Loop with Feedback Iterator (Described in IV).
7. Extract electromagnetic performance with Thermal and Structural feedback converged.

B. Results

As demo purpose, we used a 5-pole filter shown in Fig. 3. Ceramic dielectric loss tangent and aluminum conductivity have a thermal dependency. Thermal and structural loads are applied to Ansys HFSS, predicting the degraded electromagnetic performance of the filter. Fig. 4 shows the S-Parameter degradation and frequency shift of 8 MHz due to the Multiphysics coupling.

To validate the entire flow, the simulation results were compared with laboratory measurements.

Fig. 5 shows the comparison between measurements and simulation of the filter zero frequency shift for initial temperatures from 25ºC to 65ºC. Solid line represents the simulation results and the dashed line represents the measurements. As we can see, the plot shows an excellent agreement between simulation and measurements.

VI. Conclusions

This paper illustrated a reliable simulation workflow able to analyze thermo-electro-mechanical stress effects within filters. This workflow allows engineers to prevent failures during the early stages of the design process and to reduce time-to-market. All the steps of the finite element analysis in the different physics are automated and have been performed with Ansys tools and PyAEDT library.

Finally, we showed how the multiphysic coupling can predict the electromagnetics performance distortion in the device in both insertion losses and in the resonance frequency.

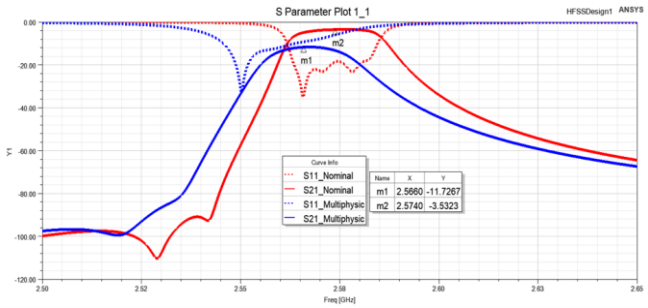


Fig. 4 - Filter S-Parameters. Red line: nominal; Blue line: Multiphysic coupling

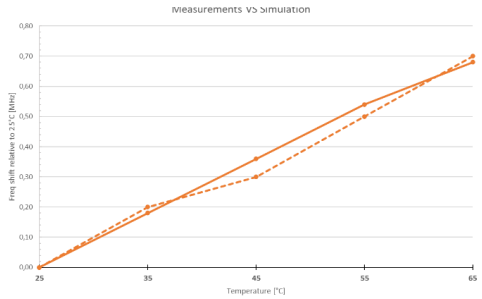


Fig. 5 - Comparison between measurements and simulation of the filter zero frequency shift. Solid line: simulation; dashed line: measurements

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